SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Total Marks: 50

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Code of Conduct

I G.Venkata Goutham Reddy (192324265) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Goutham

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks:

Cloud computing fundamentals

CLOUD COMPUTING
Delivery of computing services over the Internet.

is characterized by / provides / has

Characteristics
- on-demand access
- Resource pooling
- Elasticity.
- Measured service
- Broad network access

Benefits
- Lower initial cost
- scalability
- Flexibility
- Anywhere access
- Easy maintenance

Limitations
- Internet dependency
- security risks.
- possible downtime
- vendor lock-in
- unpredictable costs

service Types.
- IaaS
- PaaS
- SaaS
- XaaS

Linking logic:- cloud computing is characterized by essential features, provides benefits, has limitations, and offers different service models.

2) Evolution of cloud computing.

EVOLUTION OF CLOUD COMPUTING

developed through

1. Hardware Evolution

↑mainframes → personal computers → powerful servers.

enabled

2. Distributed Systems

connected computers share processing and resources.

combined with

3. Internet- and software Evolution.

Internet- → web applications → APIs and web services

led to

4. cloud Services Delivery

IaaS → PaaS → SaaS

on-demand services over the Internet

Linking logic!- Hardware advances and distributed systems, combined with internet- and software development, enabled modren cloud service delivery.

Cloud service models.

CLOUD SERVICE MODELS

are classified into

IaaS – Infrastructure as a Service
provider:- servers, storage, networking, virtualization.
User:- OS, application, data, configrations.

PaaS – Platform as a service.
provider:- Infrastructure, OS, runtime, development-platform.
User:- Application and data.

SaaS – Software as a service.
provider:- Application and underlying infrastructure.
User:- uses the software and manages their data and access.

XaaS – Everything as a service.
provider:- Delivers a specific IT resource or capability as a service.
user:- consumes and configures the service as permitted.

Linking logic:- Service models determine what the provider manages and what the user manages. control generally shifts from the user toward the provider from IaaS to SaaS

4) Virtualization and web services as Enablers of cloud.

CLOUD COMPUTING

is enabled by

Virtualization
- uses a Hypervisor.
- creates virtual machines.
- provides Abstraction
- enables resources pooling and rapid provisioning

Web services.
- use APIS
- include SOAP and REST
- connect provider and consumer
- enable automated access and integration.

together support

Cloud deployment and scalability.

Elasticity. on-demand provisioning. Resource sharing

- Automated management

Linking logic:-

- virtualization abstracts physical hardware and allows resources to be allocated flexibly.
- web services use APIs, SOAP, and REST to enable communication between systems.
- the provider supplies cloud resources, while the consumer accesses them through services and interfaces.
- Together, they support rapid deployment, scalability and elasticity.

12/10/26